

01 / WHAT IT IS • WHO IT IS FOR

An investigation assistant between experiments

The plasma is colder than expected.
More heat lost, or less heating delivered?

HEATING

Commanded power

How much reached the plasma?



TRANSPORT

Heat leaving the core

Did transport change?

For physicists and diagnostic teams: decide what to measure next.

What can the investigator see?

A research question + operator-configured site tools

SITE ANALYSIS

Profiles and parameters

Temperature, density,
heating and device geometry

Synthetic reduced 1-D model.
Full arrays stay in the analysis code.



INVESTIGATOR MODEL

Released products

Context, transport category,
11 rounded profile samples,
source-audit finding or refusal

Fixed products with evidence IDs.



RESEARCHER

Report + tool record

Evidence citations
Missing information
Recommended measurement

Inspect the report against the
actual tool results.

Example recommendation: **independently verify delivered heating at A.**

Why use an agent here?

The question can change

A missing audit or unsuitable comparison can change the next useful request.

A fixed workflow remains the baseline. Pooling is an option when permitted.

5 units per site / case

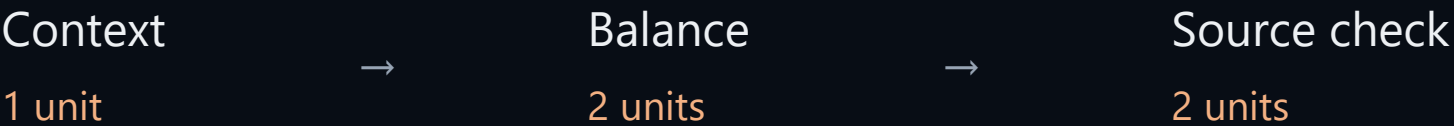
Repeat release → cache

Steward approval → gateway still checks

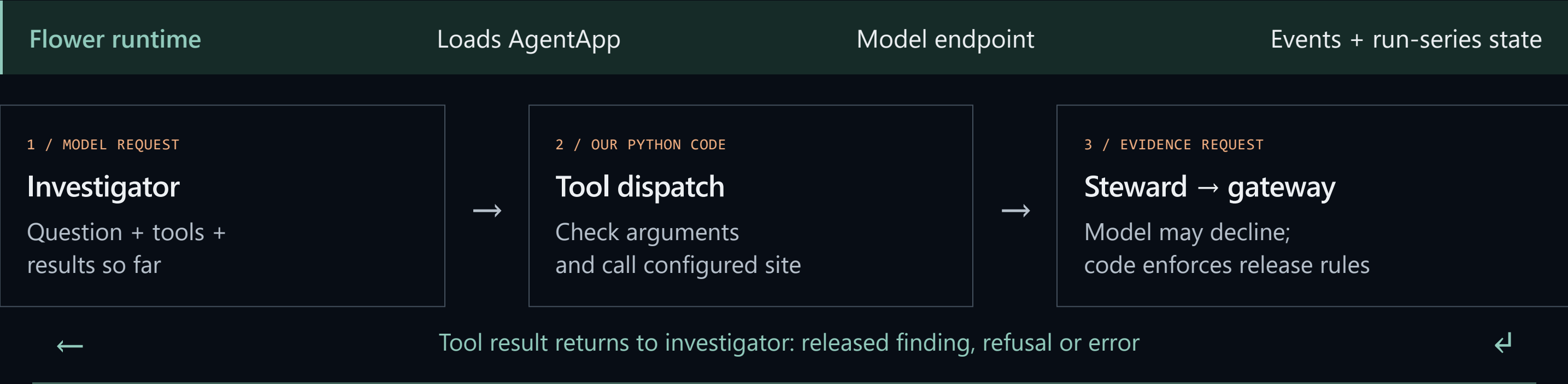
Disclosure accounting, not differential privacy. Agent benefit still needs evaluation.

The release rules stay in code

The model requests a product.
The gateway decides whether to release it.



Inside one Flower AgentApp run



STOP AND REPORT

Up to 8 planning turns / 16 tool calls by default.
Final report: tools disabled → events emitted to Flower.

TWO SEPARATE STORES

Flower Context: app-defined audit for the run series.
Gateway SQLite: releases, costs and cached evidence.

Our app owns the loop, tools and site adapters. **Default sites share one process.**

What happens in the heating example?

Captured gateway results · synthetic facilities · fixed requests, no LLM

- A

Apparent transport is elevated

Source unverified; no independent audit available.

Cause unresolved
- B

Independent source audit available

Delivered power is lower; corrected transport is near reference.

Suggests A's next test
- C

Fails the demo's gradient criterion

The gateway denies its balance request.

Comparison withheld

↓

Measure delivered heating at A.

B's result motivates the measurement; A's diagnosis stays open.

Source: captured Gateway.request outputs in evidence.js

A power audit is the next step at A.

The workflow runs.

Does the agent help researchers choose better tests?

COMPARE

Agent vs. fixed workflow

Same tools and disclosure allowance.
Cases held out from development.

MEASURE

Useful diagnostic recommendations

Evidence support, policy violations,
elapsed time and model cost.

Today: executable prototype + inspectable evidence. **Next:** measured research benefit.